



CivicVoice: An AI-Driven Public Grievance Reporting and Intelligent Complaint Management System

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Transforming Governance Through Intelligent Automation

The Challenge

Traditional grievance systems rely on manual processes, physical visits, and fragmented communication—leading to delays, misclassification, and reduced public trust.

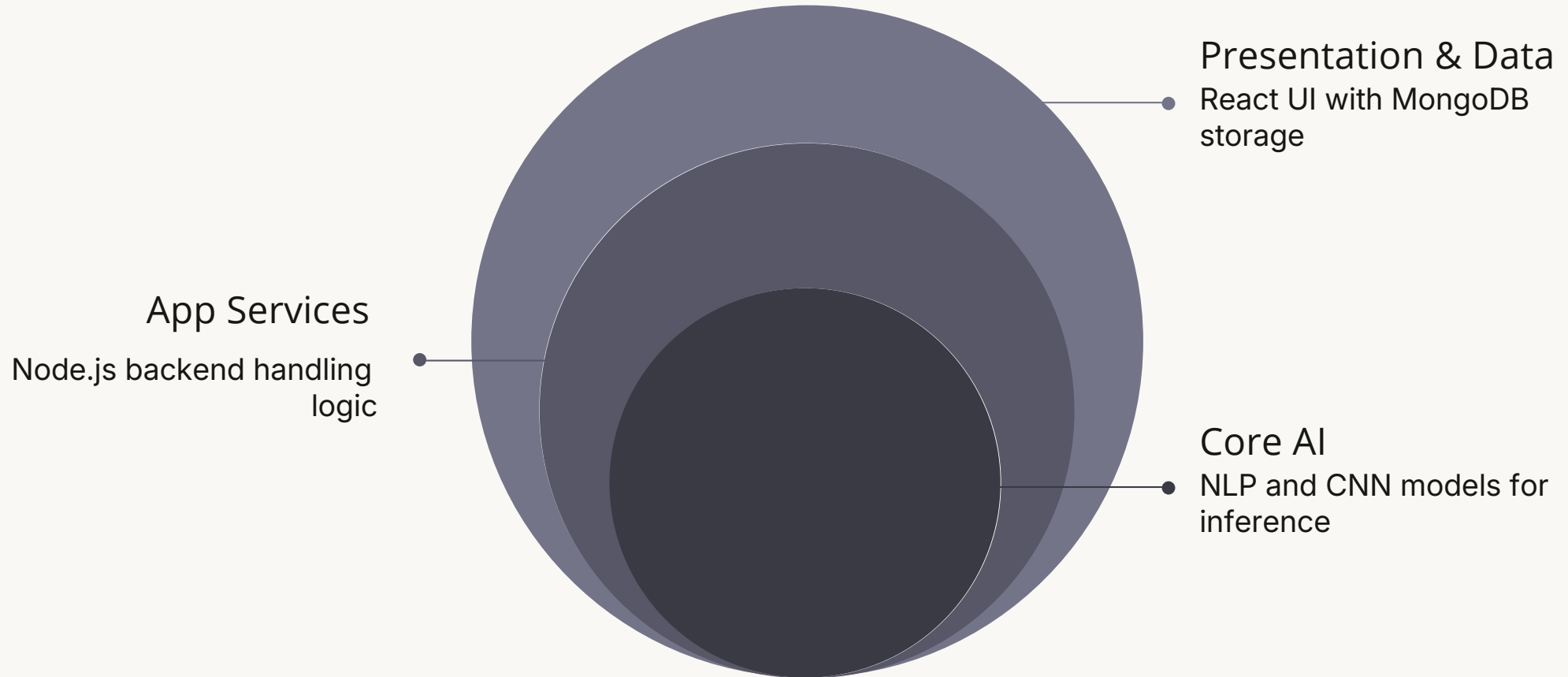
This web-based platform enables citizens to report infrastructure issues while providing administrators with real-time analytics for improved resource allocation.

The Solution

CivicVoice automates complaint classification, prioritization, and routing using AI techniques—NLP, CNN-based image analysis, and sentiment analysis.



System Architecture Overview



Core System Components



Complaint Submission

Structured digital forms with React.js interface, image uploads, and real-time validation



Authentication

JWT-based secure token system with role-based access control



Routing Engine

Automated classification and department assignment using AI analysis



Notifications

Real-time email and SMS updates on complaint status changes



Analytics Dashboard

Visual metrics on complaint trends, department performance, and geographic patterns

AI Processing Pipeline

Image Classification

Convolutional Neural Networks (MobileNet/ResNet) process uploaded images using OpenCV preprocessing. Models classify visual evidence into infrastructure categories: road damage, waste accumulation, water leakage, electrical faults.

Text Analysis

spaCy and HuggingFace Transformers preprocess complaint descriptions through tokenization and normalization, then classify using trained models recognizing issue patterns in citizen reports.

Sentiment & Priority

Pretrained sentiment models assign urgency scores based on emotional tone and language intensity—flagging critical safety issues for immediate attention.



Implementation Workflow



The diagram illustrates a four-step implementation workflow. Each step is represented by a horizontal arrow pointing to the right, with the text of the step inside the arrow. The arrows are stacked vertically and increase in length from top to bottom, creating a sense of progression. The colors of the arrows transition from a dark charcoal grey for the first step to a medium grey for the last step.

User Login

Submit Complaint

AI Analysis

Routing & Tracking

Implementation Progress & Achievements

100%

System Architecture
Complete four-layer design
implemented with React.js frontend,
Node.js backend, MongoDB storage,
and integrated AI modules

100%

Authentication, complaint submission,
routing engine, notification system, and
analytics dashboard fully operational

4

AI Components
NLP text classification, CNN image
analysis, sentiment scoring, and routing
algorithm successfully integrated

Key Milestones Achieved: Database schema design completed, RESTful API endpoints implemented, user interface developed with responsive design, machine learning models trained on labeled complaint datasets, and administrative dashboard with visual analytics deployed.

Experimental Results

Performance Metrics

- **High classification accuracy** in text and image analysis modules
- **Fast processing time**—complaints classified and routed within seconds
- **Accurate routing** to appropriate municipal departments
- **Scalable architecture** handling simultaneous complaint submissions

Key Findings

Integration of AI modules

significantly reduced manual processing workload while improving speed and accuracy of complaint routing. User feedback indicated high satisfaction with tracking capabilities and notification system.



System Limitations & Challenges

Training Data Dependency

Model accuracy depends on availability of high-quality labeled datasets representing diverse complaint categories

Text Variability

Different vocabulary and phrasing by citizens can occasionally reduce NLP classification effectiveness

Image Quality

Low-quality, blurred, or poorly lit images negatively impact CNN model performance

Platform Accessibility

Current web-based system could expand to mobile applications for broader citizen reach

Future Enhancements & Conclusion

- **Mobile applications** for Android and iOS platforms
- **Multilingual support** for regional language submissions
- **IoT integration** with sensors detecting infrastructure failures automatically
- **Advanced deep learning** using large language models for improved accuracy

CivicVoice demonstrates a scalable, intelligent approach to public grievance management—transforming traditional systems into transparent, efficient platforms aligned with smart city development goals.

